

L_RAG: A Controlled Ablation Study of Vietnamese Legal Retrieval-Augmented Generation

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Abstract

This report evaluates design choices in L_RAG, a Vietnamese legal retrieval-augmented generation system, on a fixed 500-question benchmark (400 answerable and 100 unanswerable questions). We report four controlled ablation families rather than a pooled comparison: embedded chunk text, dense versus hybrid retrieval with graph expansion, reranking within a shared candidate pool, and Base versus chain-of-thought (CoT) prompting. Retrieval is scored against gold chunk identifiers; lexical answer overlap, template-based answerability behavior, structural citation fields, and recorded latency are separate diagnostics. Metadata improves document success at depth 10 by 4.5 percentage points but reduces MRR. The tested BM25+RRF hybrid loses 5.98 Recall@10 points to dense retrieval and increases median total latency from 5.483 to 31.538 seconds. One-hop graph expansion loses 11.81–12.40 Recall@10 points, chiefly by replacing seed evidence. In a common 30-candidate pool, Jina reranking improves Recall@10 by 9.23 points and nDCG@10 by 11.85 points over RRF-only. Under fixed context, CoT decreases GPT-4o-mini Token F1 by 1.57 points. These are benchmark-bounded engineering findings: lexical and structural diagnostics do not establish legal correctness, citation entailment, or faithfulness.

1 Introduction

Guidance: Introduce the Vietnamese legal QA problem, motivate traceable evidence, and preview the report contributions. Keep this chapter focused on why the problem matters; reserve implementation details for Sections 4 and 5.

1.1 Background and Motivation

Guidance: Add a motivating Vietnamese legal question, the governing provision, and an explanation of why a cited answer is required. Include the

planned motivation figure and the keyword-search versus vector-RAG versus L_RAG contrast.

1.2 Problem Statement

Guidance: State the failure modes to be addressed: long documents, hierarchy blindness, cross-document references, outdated provisions, missing text, and unsupported answers. Add the problem-to-consequence table.

1.3 Project Objectives

Guidance: List measurable objectives and map each objective to an implemented module, source file or script, and validation evidence. Add the contribution map.

1.4 Scope of the System

Guidance: Define the included corpus, preprocessing, graph, retrieval, generation, and evaluation stages. Explicitly list excluded capabilities and the assumptions that limit deployment.

1.5 Main Contributions

Guidance: Summarize the structured legal corpus, G-LRAG graph, hybrid retrieval pipeline, citation-grounded generation, and controlled ablation study. Use one contribution overview figure.

1.6 Report Organization

Guidance: Explain the Problem → Data → Methodology → Implementation → Evaluation → Ablations → Discussion → Conclusion flow and add the report-roadmap figure.

2 Background and Related Work

Guidance: Replace these prompts with a focused literature review. Explain which prior ideas motivate each L_RAG design choice and identify the gap addressed by the controlled ablations.

2.1 Retrieval-Augmented Generation

Guidance: Define the standard query, retriever, context, prompt, LLM, and answer pipeline. Add a standard RAG diagram and a grounding illustration.

2.2 Legal Information Retrieval

Guidance: Discuss authority, hierarchy, temporal validity, citation dependencies, and the cost of retrieval errors in legal settings. Add the legal-document hierarchy and a legal retrieval example.

2.3 Dense, Sparse, and Hybrid Retrieval

Guidance: Compare dense retrieval, BM25, and rank fusion. Add the retrieval comparison table and a vocabulary-mismatch example.

2.4 Knowledge-Graph-Augmented Retrieval

Guidance: Explain how graph structure supplies containment, neighboring, citation, amendment, and validity context. Add a graph-retrieval illustration and a vector-versus-graph comparison table.

2.5 Reranking and Citation-Grounded Generation

Guidance: Introduce candidate reranking, cross-encoders, citation grounding, and abstention. Add the two-stage ranking and citation-grounding illustrations.

2.6 Design Requirements for Legal RAG

Guidance: Convert legal reliability requirements into design choices and evaluation checks. Add a requirement traceability matrix and design rationale diagram.

3 Legal Dataset and Benchmark Construction

Guidance: Describe the corpus and benchmark as reproducible data products. Keep the running Vietnamese legal question consistent across the dataset, graph, retrieval, and generation examples.

3.1 Legal Corpus and Raw Data Sources

Guidance: Inventory raw files, record types, identifiers, relationships, and text fields. Reuse the document-type distribution and add a sanitized metadata and relationship example.

3.2 Dataset Normalization and Cleaning

Guidance: Document identifier, date, document-type, status, and relationship transformations. Reuse the data-pipeline figure and add a before-and-after record plus a normalization-rules table.

3.3 Relationship Processing and External References

Guidance: Explain resolved in-corpus targets, unresolved external targets, external stubs, relationship vocabulary, and direction verification. Add the relationship-resolution diagram, vocabulary table, and stub example.

3.4 Quarantine and Data-Quality Verification

Guidance: Show accepted records, quarantined records, verification gates, final inclusion, audit counts, and reconciliation identities. Reuse the edge-quarantine asset.

3.5 Legal Text Structuring

Guidance: Describe the Document → Provision → Chunk hierarchy and provenance pointers. Reuse the provision and chunk assets and add the structured-artifact inventory.

3.6 QA Benchmark Construction

Guidance: Show context sampling, question generation, verification, diversity filtering, deduplication, and final export. Add the benchmark-flow figure, a record example, and stage-count table.

3.7 Question Categories and Answer Types

Guidance: Define single-hop, citation, legal-validity, multi-hop, and cross-document categories, and relate them to abstractive, boolean, extractive, and unanswerable answer types. Add the category-answer-type matrix.

3.8 Ground-Truth Evidence

Guidance: Explain reference answers and document, provision, and chunk identifiers. Add the evidence-link diagram, annotated JSONL record, and ground-truth coverage table.

3.9 Dataset Statistics and Validation

Guidance: Report corpus size, missing-text counts, orphan counts, quarantine counts, external stubs, and reconciliation checks. Reuse the length, node-type, and document-type assets and add a validation dashboard.

4 L_RAG System Design and Methodology

Guidance: Explain how the data products become an end-to-end legal RAG system. Use the same identifiers and running example introduced in Section 3.

4.1 End-to-End System Architecture

Guidance: Reuse the system-architecture asset and add a labeled version covering corpus construction, graph construction, vector indexing, retrieval, ranking, generation, and output. Include a component responsibilities table.

4.2 Offline Data and Index Construction

Guidance: Describe the normalized-data branches for structured text, graph artifacts, and vector-index artifacts. Reuse the graph-pipeline asset and add the offline pipeline and artifact-lineage diagrams.

4.3 Online Query-Processing Workflow

Guidance: Add a sequence diagram for query embedding, dense/BM25 retrieval, graph filtering or expansion, fusion, reranking, context construction, and generation. Include decision paths for insufficient evidence, invalid filters, and unanswerable questions.

4.4 Knowledge Graph Construction

Guidance: Define Document, ExternalStub, Provision, and Chunk nodes; structural and legal edges; validity and authority overlays; traversal; and the graph API mapping. Reuse the graph-pipeline, node-distribution, and edge-dashboard assets.

4.5 Retrieval and Ranking

Guidance: Explain chunk-text construction, E5 and FAISS retrieval, Vietnamese BM25, RRF, graph expansion, and the 30-candidate reranker pool. Add the planned diagrams and reranker-configuration table.

4.6 Answer Generation

Guidance: Describe context serialization, prompt anatomy, answer formats, citation mapping, and the insufficient-context branch. Do not include hidden chain-of-thought; show only observable prompt and output structure.

4.7 End-to-End Worked Example

Guidance: Carry one real benchmark question through source document, structured provision, chunk, graph neighborhood, ranked evidence, and final cited answer. Add the six-stage storyboard, evidence table, and success or failure comparison.

5 Implementation and Experimental Setup

This section states the controls and scoring rules needed to interpret the ablations in Section 6. The experiment families share a benchmark but are not a single factorial experiment: candidate pools, context construction, and latency accounting differ. Variants are therefore compared only within their own controlled family.

5.1 Pipeline and Frozen Artifacts

The evaluated pipeline has five stages: query embedding with `intfloat/multilingual-e5-large`; dense FAISS retrieval and, where enabled, tokenized BM25 retrieval; RRF fusion, graph expansion, or cross-encoder reranking; serialization of retained chunks into a grounded prompt; and answer generation and evaluation. The analysis reads frozen manifests, predictions, aggregate metrics, latency records, and error logs; it does not rerun models. Every primary comparison is joined by `qa_id`, so an aggregate difference is traceable to paired cases.

5.2 Indexes, Graph, and Runtime Settings

The dense/hybrid/graph experiment uses a FAISS index with 1,513,376 chunk vectors and a graph over the same number of chunks. The hybrid branch searches ten BM25 shards. Dense/hybrid retrieval uses a 50-result search budget, broad filtering, a final context of 10 chunks, and RRF parameter $k = 60$ where BM25 is enabled. Graph variants take five seeds, traverse one hop, and retain at most 10 graph-context chunks. The dense/hybrid and reranker generation artifacts use GPT-4o-mini.

The reranker family fixes a 30-chunk RRF candidate pool (C30) and a 10-chunk final context (T10), comparing no reranking, RRF-only, mMiniLM, Qwen3, BGE, and Jina. The primary generator study compares GPT-4o-mini Base and CoT with the saved top-10 context fixed by ques-

tion. The metadata study changes only the indexed string from chunk body to header-plus-chunk text; encoder, FAISS, top- $k = 10$, generator, prompt, seed, corpus, and benchmark stay fixed.

5.3 Benchmark and Metrics

The benchmark contains 500 questions: 400 answerable questions with gold chunk identifiers and 100 unanswerable questions. Retrieval scores use only the 400 answerable records. Given gold chunks G and retrieved top- k chunks R_k , context recall is $|G \cap R_k|/|G|$; Hit@ k indicates that at least one gold chunk occurs. MRR and nDCG reward early gold rank. Candidate Recall@30 is the evidence ceiling for the reranker study and is intentionally identical for every reranker.

Exact Match, Token F1, and ROUGE-L are lexical overlap with reference answers. The saved unanswerable score is a refusal-marker diagnostic, not a semantic assessment of legal appropriateness. Citation presence and structural coverage measure saved marker/parser fields, not claim-to-source entailment. Latency is reported as saved p50/p95 timing; cached-BM25 reranker timing is not full on-line sparse-retrieval timing.

5.4 Pairing, Uncertainty, and Claim Boundaries

Headline effects are paired by qa_id. Retrieval uses the shared answerable intersection; scheduled-denominator answerability diagnostics keep all 500 cases. Percentile paired-bootstrap confidence intervals use 10,000 resamples for headline metadata, dense/hybrid, reranker, and GPT comparisons; reranker slices use 2,000. Intervals crossing zero are reported as inconclusive. The non-GPT prompt pairs are descriptive when code provenance, timeouts, retries, or retrieved context differ.

No blinded legal-correctness labels, claim-level faithfulness labels, or validated semantic judge are available. Thus, no result is interpreted as proof of legal correctness, legal safety, citation entailment, or faithful reasoning merely because it improves a lexical, rank, citation-format, or refusal-format metric.

6 Ablation Studies

This chapter follows the ablation plan in report_structure.md. We report five completed experiment families and retain the deterministic evidence available for each one.

Comparisons are made within a family because candidate construction and context budgets differ; the results are not treated as one pooled factorial experiment.

6.1 Embedding Text Ablation: Chunk-Only versus Chunk+Metadata

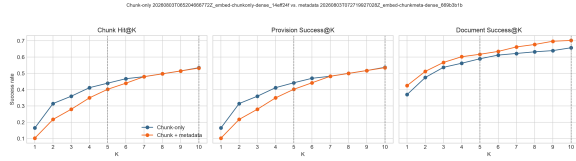
This comparison holds the dense encoder, generator, benchmark, seed, and raw FAISS evaluation path fixed. Only the indexed text changes from the chunk body to a header-plus-chunk representation. On 400 paired answerable questions, metadata leaves exact chunk Recall@10 effectively unchanged (48.5% versus 48.5%) but increases document-level success from 65.8% to 70.2%. The paired document-success gain is +4.5 percentage points (95% CI [+0.5,+8.8]), whereas the MRR@10 change is −5.9 points (95% CI [−9.1, −2.7]). Metadata therefore improves document identification while making the first exact gold chunk appear later; the complete scores are in Table 1 and the retrieval curves and paired outcomes are shown in Figure 1.

6.2 Retrieval Ablation: Dense versus Hybrid Retrieval

The dense/hybrid comparison uses the same header-plus-chunk dense index, 50-seed retrieval budget, 10-chunk final context, benchmark, and generator. Hybrid retrieval adds BM25 and reciprocal rank fusion. DenseOnly outperforms HybridOnly on Recall@10 (48.55% versus 42.57%) and nDCG@10 (0.2813 versus 0.2638); the full comparison is in Table 2. The paired Recall@10 difference is −5.98 points with a 95% CI of [−9.71, −2.21]. Hybrid retrieval also raises recorded total median latency from 5.483 seconds to 31.538 seconds; paired uncertainty is reported in Table 3.

6.3 Graph Ablation: With versus Without Graph Expansion

Graph variants expand five seed chunks by one hop and retain at most 10 chunks. Graph expansion currently displaces more evidence than it adds. Relative to DenseOnly, DenseGraph changes Recall@10 by −12.40 points and loses at least one previously retrieved gold chunk for 61 questions while adding a graph-only gold chunk for only one question. The HybridOnly to HybridGraph change is −11.81 points, with 59 questions losing



Chunk-only 20260803T0652046667722_embed-chunkonly-dense_14eff24f vs. metadata 20260803T0727190270282_embed-chunkmeta-dense_669b3b1b

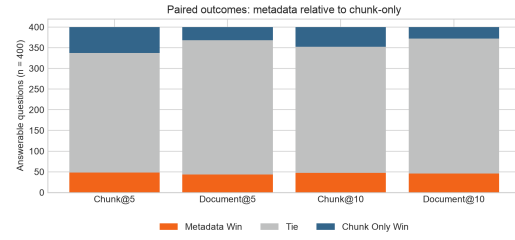


Figure 1: Embedding-text ablation diagnostics. The left panel shows success curves by retrieval depth; the right panel shows paired win, tie, and loss counts. Both plots use the shared answerable-question comparison.

Table 1: Embedding-text ablation on 400 paired answerable questions. Values are percentages; document success means that a retrieved chunk from the gold document is present in the indicated context. Deltas are chunk+metadata minus chunk-only.

Variant	Hit@5	Recall@5	MRR@5	Doc.@5	Hit@10	Recall@10	MRR@10	Doc.@10
Chunk-only	44.0	39.0	27.4	59.0	53.5	48.5	28.6	65.8
Chunk+metadata	40.2	36.8	20.9	61.8	53.2	48.5	22.7	70.2
Delta	-3.8	-2.2	-6.5	+2.8	-0.2	+0.1	-5.9	+4.5

a gold chunk and six gaining one; the displacement counts and rank deltas are summarized in Table 4.

Graph contexts are denser because they contain fewer chunks on average (5.605 for DenseGraph and 6.305 for HybridGraph), and their identity precision is higher. This is a precision–coverage trade-off, not evidence of an overall retrieval-quality improvement.

6.4 Reranker Ablation

The reranker family compares no reranking, RRF-only, mMiniLM, Qwen3, BGE, and Jina over a reconstructed shared RRF pool of 30 candidates and a final top 10. All six runs cover the same 500 questions without retrieval or generation errors, and the reconstruction matches the recorded RRF-only top-10 prefix for 500/500 questions. Candidate Recall@30 is 0.5995 and Candidate Hit@30 is 0.6450 for every configuration, so the comparison measures ranking and retention within a fixed evidence ceiling.

Jina is the strongest retrieval configuration: relative to RRF-only, it improves Recall@10 by +9.23 points, MRR@10 by +13.35 points, and nDCG@10 by +11.85 points; all three 95% paired-bootstrap intervals exclude zero. Qwen3 and BGE also have positive intervals, while mMiniLM supplies a smaller positive gain at the lowest reranker cost. BGE has the highest token F1 and ROUGE-L, whereas Jina has the highest observed unanswerable accuracy. These answer metrics are lex-

ical and abstention diagnostics, not legal correctness judgments. Aggregate scores and latency are linked in Table 5, with paired effects in Table 6 and diagnostic curves in Figure 2.

6.5 Generator Ablation: Base Prompt versus Chain-of-Thought Prompt

The generator-reasoning family keeps the intended retrieved top-10 context fixed for each question and changes the prompt mode. The primary comparison uses complete GPT-4o-mini Base and CoT runs: both cover all 500 benchmark questions and share 400 answerable question IDs. The frozen artifacts report zero retrieval-context mismatches. The DeepSeek-V3.1-Thinking, GLM-5, and Qwen3-8B pairs are complete but descriptive; four shared DeepSeek cases have different exact top-10 chunk-ID sequences, and the Qwen pair uses different code snapshots, timeout values, and retry limits.

On the 400 paired answerable questions, CoT decreases Token F1 from 24.72% to 23.15% (−1.57 points) and ROUGE-L from 21.60% to 19.90% (−1.71 points). Exact match is essentially unchanged (14.00% to 13.75%, −0.25 points). The Token F1 win/tie/loss count is 79/164/157 in favor of Base. The run inventory is in Table 7 and the paired GPT-4o-mini result is linked in Table 8.

Table 2: Dense versus hybrid retrieval. Retrieval values use 400 answerable questions; latency percentiles use all 500 cases.

Method	Recall@5	Recall@10	MRR@10	nDCG@10	Retrieval p50/p95	Total p50/p95
DenseOnly	0.3678	0.4855	0.2267	0.2813	0.797/0.856	5.483/9.946
HybridOnly	0.3140	0.4257	0.2252	0.2638	27.855/27.924	31.538/33.623

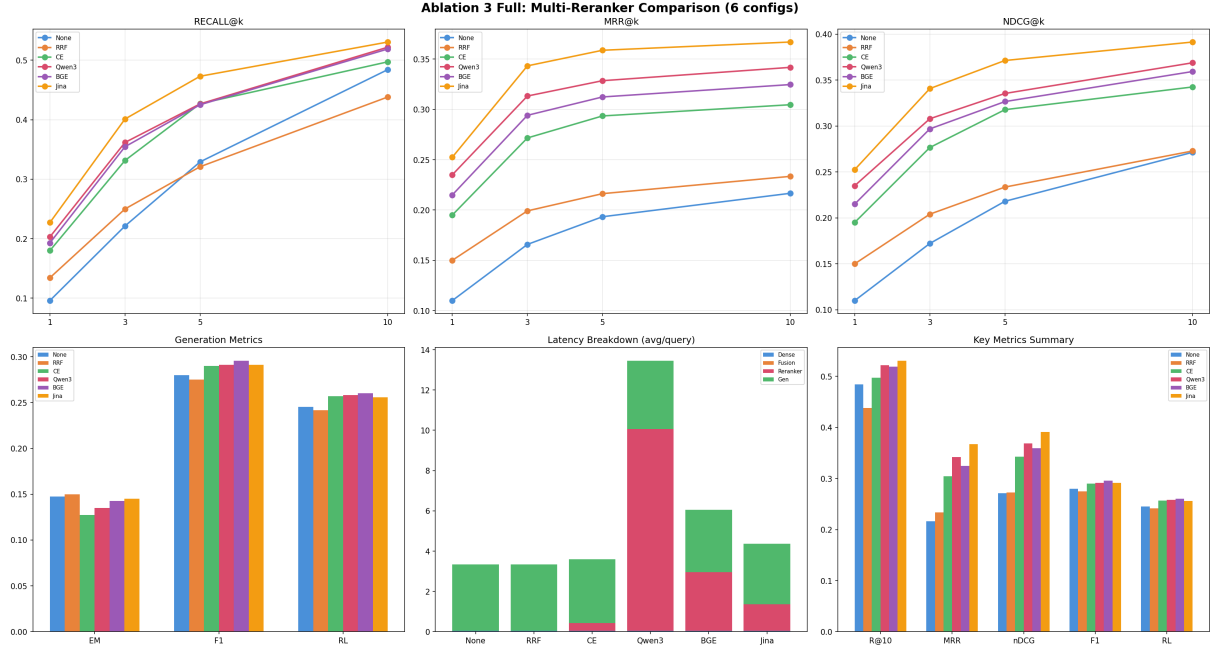


Figure 2: Reranker curves, answer-overlap diagnostics, latency breakdown, and headline metrics from the six-configuration run. The paired confidence intervals in Table 6 are the primary uncertainty analysis.

Table 3: Paired hybrid-minus-dense differences.

Metric	Mean delta	95% CI
Recall@10	-0.0598	[-0.0971, -0.0221]
MRR@10	-0.0015	[-0.0299, +0.0271]
nDCG@10	-0.0175	[-0.0437, +0.0085]

6.6 Performance by Question Category and Answer Type

The largest Jina gains occur for legal-validity and citation questions, while the smaller extractive slice is inconclusive. The multi-hop slice has only 20 questions and is exploratory; hard and cross-document slices are too small for a main conclusion. The table reports Jina versus RRF-only deltas on answerable cases with 2,000 paired-bootstrap resamples; see Table 9.

6.7 Evidence Coverage, Abstention, and Citation Analysis

The GPT-4o-mini pair provides structural diagnostics for evidence availability, answerability decisions, and citations. These fields describe observ-

able output format and saved parser coverage; they do not establish citation entailment, semantic refusal quality, or legal correctness. CoT lowers the false-refusal rate on answerable questions from 21.25% to 19.50%, while unanswerable recognition remains 98.00% for both prompts. Citation presence rises from 78.25% to 81.00% and mean unique sources from 1.46 to 1.68, but structural citation coverage falls from 89.42% to 87.91%; the detailed diagnostics are in Table 10.

Evidence-stratified deltas do not reverse the result: CoT Token F1 is lower when all gold support is retrieved (-1.98 points, $n = 138$), when support is partial (-1.86 points, $n = 29$), and when support is absent (-1.30 points, $n = 233$). The saved records cannot support claim-level faithfulness.

6.8 Latency and Efficiency Comparison

Efficiency differences are substantial. Hybrid retrieval reports a one-time BM25 precomputation cost of about 13,495 seconds and a much larger per-query retrieval cost. Among rerankers,

Table 4: Effect of graph expansion on top-10 gold evidence. Counts are over 400 answerable questions; graph-only gold is introduced by expansion and graph-missed gold was present before expansion but absent afterward.

Base	Δ Recall@10	Δ MRR@10	Δ nDCG@10	Graph-only gold	Graph-missed gold	Avg. chunks	ID precision
DenseOnly $\rightarrow$ DenseGraph	-0.1240	-0.0222	-0.0439	1 query (0.25%)	61 queries (15.25%)	5.605	0.0800
HybridOnly $\rightarrow$ HybridGraph	-0.1181	-0.0240	-0.0411	6 queries (1.50%)	59 queries (14.75%)	6.305	0.0641

Table 5: Multi-reranker ablation over a shared 30-candidate pool and final top-10 context. Retrieval and overlap values are means over 400 answerable questions; recorded unanswerable accuracy and p50 latency use the complete 500-case runs.

Model	Recall@10	Hit@10	MRR@10	nDCG@10	Token F1	ROUGE-L	Unans. acc.	Rerank p50	Total p50
None	0.4842	0.5375	0.2167	0.2714	0.2801	0.2453	0.856	0.000	2.925
RRF-only	0.4382	0.4800	0.2335	0.2728	0.2750	0.2416	0.874	0.000	2.972
mMiniLM	0.4973	0.5425	0.3047	0.3425	0.2901	0.2568	0.872	0.385	3.343
Qwen3	0.5218	0.5725	0.3418	0.3688	0.2912	0.2583	0.876	9.695	13.007
BGE	0.5188	0.5675	0.3247	0.3593	0.2955	0.2603	0.868	2.872	5.848
Jina	0.5305	0.5850	0.3670	0.3914	0.2913	0.2558	0.884	1.278	4.158

mMiniLM is fastest, Jina is intermediate, BGE is slower, and Qwen3 is much slower. The generator CoT prompt raises generation p50/p95 from 2.92/5.74 seconds to 3.68/6.17 seconds, and total p50/p95 from 13.99/17.21 to 14.74/18.42 seconds. Reranker p95, peak GPU memory, and controlled full-online throughput are unavailable. The complete stage and end-to-end comparison is linked in [Table 12](#).

6.9 Qualitative Ablation Examples and Failure Cases

The saved aggregate artifacts support a precise failure taxonomy even without hand-labelled legal case studies. Metadata has a document-provision trade-off: document-level success increases at top 10 while exact chunk MRR falls, so a retrieved document title is not proof that the governing provision is ranked early. Hybrid fusion can add lexical context while lowering exact gold-chunk coverage; a generation-overlap gain is not evidence that RRF preserved the gold ranked list. Graph expansion has a direct seed-displacement failure: it introduces a new top-10 gold chunk for only 1/400 Dense and 6/400 Hybrid questions, but removes at least one seed-retrieved gold chunk for 61 and 59 questions respectively. Finally, a fluent response with a structural citation marker is not a verified legally correct or entailed answer.

Future case studies should preserve the qa_id, gold chunk IDs, ordered candidate IDs before and after the changed component, serialized context, final answer, and citation parser output. A case must label whether its diagnosis is an automatic rank fact or a human legal judgement; this keeps

illustrative examples auditable rather than anecdotal.

6.10 Summary of Ablation Findings

The completed evidence supports conditional engineering decisions. Metadata helps document-level retrieval but not exact chunk ranking. The tested hybrid configuration reduces exact-chunk coverage and adds latency. One-hop graph expansion increases context density while displacing more gold evidence than it adds. Cross-encoder reranking is the clearest positive retrieval result: Jina gives the strongest ranking scores, BGE gives the strongest lexical overlap, and mMiniLM gives the lowest recorded reranking cost. For the primary generator pair, CoT lowers lexical overlap under fixed context while improving only template-based decision diagnostics and adding latency. The consolidated recommendations are linked in [Table 13](#).

Table 6: Paired effects of each reranker relative to RRF-only. Deltas are in percentage points; intervals are 95% percentile bootstrap CIs from 10,000 question-level resamples over the 400 answerable cases. A negative false refusal delta indicates fewer refusals on answerable questions.

Reranker	Δ Recall@10	Δ MRR@10	Δ nDCG@10	Δ false refusal
mMiniLM	+5.91 [+1.85, +9.96]	+7.12 [+3.55, +10.68]	+6.96 [+3.77, +10.26]	+0.25 [−3.00, +3.50]
Qwen3	+8.37 [+4.57, +12.25]	+10.84 [+7.24, +14.36]	+9.60 [+6.52, +12.73]	−0.25 [−3.25, +2.75]
BGE	+8.07 [+4.29, +11.80]	+9.12 [+5.50, +12.70]	+8.65 [+5.47, +11.84]	+0.75 [−2.25, +3.75]
Jina	+9.23 [+5.51, +12.98]	+13.35 [+9.51, +17.24]	+11.85 [+8.46, +15.27]	−1.25 [−4.00, +1.25]

Table 7: End-to-end generator-reasoning run inventory. Lexical metrics are means over successful answerable cases; decision accuracy is the saved template-based answerability diagnostic.

Model	Prompt	Run directory	Valid/500	EM (%)	Token F1 (%)	ROUGE-L (%)	Decision acc. (%)
GPT-4o-mini	Base	4o-mini-base	500/500	14.00	24.72	21.60	82.60
GPT-4o-mini	CoT	4o-mini-CoT	500/500	13.75	23.15	19.90	84.00
DeepSeek-V3.1-Thinking	Base	deepseek-base	500/500	22.25	17.84	15.28	92.00
DeepSeek-V3.1-Thinking	CoT	deepseek-CoT	500/500	22.25	16.51	14.12	92.20
GLM-5	Base	glm5-base	500/500	25.25	16.91	14.90	91.80
GLM-5	CoT	glm5-CoT	500/500	25.00	16.23	14.27	93.40
Qwen3-8B	Base	qwen3-8b-base	500/500	19.25	21.98	18.92	95.80
Qwen3-8B	CoT	qwen3-8b-CoT	500/500	17.25	20.05	17.17	95.20

Table 8: Primary paired GPT-4o-mini Base-versus-CoT comparison on 400 answerable questions. Deltas are CoT minus Base in percentage points; CIs are percentile bootstrap intervals from 10,000 paired resamples.

Metric	Base (%)	CoT (%)	Δ (pp)	95% paired CI (pp)
Exact Match	14.00	13.75	−0.25	[−1.50, +1.00]
Token F1	24.72	23.15	−1.57	[−2.24, −0.92]
ROUGE-L	21.60	19.90	−1.71	[−2.34, −1.10]

Table 9: Jina versus RRF-only on selected category and answer-type slices. Deltas are percentage points with 95% paired-bootstrap CIs.

Dimension	Slice	<i>n</i>	Δ Recall@10	Δ MRR@10	Δ nDCG@10
Category	citation	106	+11.89 [+2.82, +21.70]	+16.22 [+7.86, +24.51]	+15.82 [+7.51, +23.82]
Category	legal validity	91	+14.47 [+6.96, +21.43]	+18.90 [+10.28, +27.26]	+15.22 [+9.13, +21.50]
Category	single hop	172	+5.04 [−0.00, +10.37]	+9.63 [+4.34, +14.69]	+8.41 [+3.81, +13.05]
Category	multi hop	20	+7.50 [−0.83, +17.50]	+10.12 [−3.42, +24.33]	+7.95 [−0.77, +17.98]
Answer type	extractive	78	+0.13 [−8.85, +9.74]	+8.53 [−0.90, +17.39]	+7.14 [−1.33, +16.23]
Answer type	abstractive	199	+13.65 [+8.50, +18.80]	+14.74 [+9.61, +20.14]	+13.81 [+9.56, +18.71]
Answer type	boolean	123	+7.86 [+1.63, +14.50]	+14.18 [+7.64, +21.13]	+11.68 [+6.27, +17.95]

Table 10: Structural evidence, abstention, and citation diagnostics for the paired GPT-4o-mini runs.

Diagnostic	Base	CoT	Interpretation
Citation presence	78.25%	81.00%	at least one marker on answerable output
Structural coverage	89.42%	87.91%	saved factual-sentence marker share
Mean unique sources	1.46	1.68	structural source count
False refusal (answerable)	21.25%	19.50%	refusal-marker diagnostic
Unanswerable recognition	98.00%	98.00%	refusal-marker diagnostic

Table 11: Evidence-availability strata for the paired GPT-4o-mini reasoning comparison. Values are lexical means on the 400 paired answerable cases.

Evidence stratum	n	Base F1	CoT F1	Δ F1	Base ROUGE-L	CoT ROUGE-L	Δ ROUGE-L
Support fully retrieved	138	28.24	26.26	−1.98	25.64	23.34	−2.30
Partial support retrieved	29	25.62	23.76	−1.86	21.04	19.45	−1.59
No support retrieved	233	22.53	21.23	−1.30	19.29	17.92	−1.37

Table 12: Recorded latency comparisons. Values are p50/p95 seconds unless noted; reranker p95 was not recorded. Cached BM25 means reranker rows are not full-online sparse-latency measurements.

Configuration	Stage p50/p95	Total p50/p95	Stage
DenseOnly	0.797/0.856	5.483/9.946	retrieval
HybridOnly	27.855/27.924	31.538/33.623	retrieval
DenseGraph	0.798/0.860	4.998/9.750	retrieval + graph
HybridGraph	27.858/27.929	31.180/33.312	retrieval + graph
GPT-4o-mini Base	2.92/5.74	13.99/17.21	generation
GPT-4o-mini CoT	3.68/6.17	14.74/18.42	generation
mMiniLM	0.385/N/A	3.343/N/A	reranker
Qwen3	9.695/N/A	13.007/N/A	reranker
BGE	2.872/N/A	5.848/N/A	reranker
Jina	1.278/N/A	4.158/N/A	reranker

Table 13: Decision summary for the completed ablation families. Quality claims are limited to the deterministic metrics available in the saved artifacts.

Family	Changed component	Quality effect	Latency effect	Decision
Embedding	chunk text + metadata	document success +4.5 pp; MRR@10 −5.9 pp	not material in headline table	use when document identity is primary
Dense/hybrid	add BM25 + RRF	Recall@10 −5.98 pp	total p50 5.483 s → 31.538 s	do not use as default under current settings
Graph	one-hop expansion	Recall@10 −12.40 pp (Dense), −11.81 pp (Hybrid)	graph stage near dense retrieval cost	gate until seed-preserving expansion is tested
Reranker	reorder shared C30 pool	Jina: Recall +9.23 pp, MRR +13.35 pp, nDCG +11.85 pp	mMiniLM fastest; Jina p50 1.278 s	prefer Jina for retrieval quality; mMiniLM for latency
Generator	Base prompt → CoT	Token F1 −1.57 pp; ROUGE-L −1.71 pp	total p50 13.99 s → 14.74 s	retain Base for this lexical benchmark

7 Discussion, Limitations, and Ethical Considerations

7.1 Interpretation of the Main Findings

The ablations yield one clear positive retrieval intervention and several conditional or negative findings. Cross-encoder reranking improves the rank of evidence already admitted to the shared C30 pool; Jina has the strongest retrieval result, while BGE has the strongest lexical answer overlap. By contrast, the tested hybrid and graph configurations reduce exact gold-chunk coverage. These statements concern a finite benchmark and its gold evidence, not legal advice quality in deployment.

7.2 Retrieval Quality, Context Coverage, and Latency Trade-offs

DenseOnly is the preferred retrieval baseline in the dense/hybrid study: it achieves Recall@10 of 0.4855 at 0.797-second retrieval p50. HybridOnly falls to 0.4257 and takes 27.855 seconds when amortized BM25 time is included. Graph contexts appear more identity-precise because they contain only 5.605–6.305 chunks on average, but their reduced context size also removes gold coverage. The reranker result has a different interpretation: it operates after a fixed candidate ceiling, so Jina’s gain is a ranking gain rather than proof that the underlying candidate retriever finds more evidence.

7.3 Implications for Legal RAG Design

For this benchmark, use header-plus-chunk embeddings when correct document identification is

more valuable than the rank of an exact chunk; otherwise prefer the chunk-only representation. Do not deploy the current BM25+RRF or one-hop graph setup as the default retrieval path without retuning. Prefer Jina when retrieval quality justifies its 1.278-second reranking p50, or mMiniLM when its lower 0.385-second cost is more important. Retain the Base prompt for lexical answer overlap. In all cases, present citations as sources to inspect and retain human review for legal interpretation.

7.4 Dataset and Coverage Limitations

The benchmark has 500 questions, of which only 400 are answerable by stored gold chunks. The hard stratum has one answerable item, while multi-hop has 20; those slices are descriptive and cannot establish stable subgroup effects. Gold chunk IDs are an evidence target, not necessarily the full set of legally acceptable authorities. Corpus coverage, language variation, temporal validity, and external legal references may all limit transfer beyond the data snapshot.

7.5 Knowledge Graph and Validity Limitations

The graph study evaluates a specific policy—five seeds, one hop, and ten retained chunks—rather than graph information in general. Its failures point to an engineering hypothesis: expansion substitutes for seeds too early. Future graph experiments should preserve high-ranked seeds, separately budget neighbors, verify relationship direction and validity semantics, and measure whether graph-only evidence is legally useful rather than merely novel.

7.6 Retrieval and Generation Limitations

The index, BM25 tokenization, RRF parameter, candidate budgets, rerankers, generator, prompt, and final context size are fixed settings, not optimized universally. Reranker timings use cached BM25 candidates and omit full-online sparse retrieval. Lexical overlap may reward reference phrasing rather than legal adequacy. Structural citations measure marker presence or parser coverage, not the relationship between a claim and the cited authority.

7.7 Evaluation Limitations

No blinded legal-correctness labels, claim-level faithfulness labels, or validated semantic judge are

available. The report therefore does not claim that any configuration is legally more correct, safer, or more faithful solely because it improves F1, ROUGE-L, citation count, or refusal markers. Full-online p95 reranker latency, peak memory, and throughput are also absent. Non-GPT prompt pairs have provenance or control differences and remain descriptive.

7.8 Ethical Considerations and Safe Use

L_RAG should support research and source discovery, not replace a qualified legal professional. Users should inspect the retrieved authority, its date and validity, and the uncertainty of an answer. Systems should avoid unnecessary personal query data, should not obscure abstentions, and should treat well-cited generated text as a review cue rather than a legal conclusion.

8 Conclusion and Future Work

8.1 Summary of the Implemented System

L_RAG combines a Vietnamese legal chunk corpus, dense FAISS retrieval, optional BM25/RRF fusion, graph expansion, cross-encoder reranking, grounded generation, and saved per-question evaluation records. The ablation report isolates four component choices using fixed question sets and paired metrics.

8.2 Main Contributions

The empirical contribution is a traceable set of controlled decisions: the embedded text representation changes document success but not exact chunk coverage; the tested hybrid and graph variants impose measurable coverage losses; reranking within a common pool produces substantial ranking gains; and CoT does not improve lexical answer overlap under fixed context. The report also separates retrieval evidence, answer overlap, refusal formatting, citation structure, and efficiency instead of collapsing them into one score.

8.3 Lessons Learned

The central lesson is that component labels are not outcomes. Hybrid, graph, CoT, and cited output are not quality guarantees. Candidate control and paired question-level deltas reveal effects obscured by aggregate scores: graph expansion increases density while removing evidence, and rerankers can only improve what has entered C30.

Frozen manifests and complete row alignment are essential before interpreting a difference as causal.

8.4 Recommended System Configuration

For the evaluated benchmark, the recommended evidence path is dense retrieval with the Jina reranker when its latency is acceptable, a 10-chunk final context, and the Base prompt. Use the graph branch only after seed-preserving expansion is validated; do not select the current hybrid branch as a default. This is a benchmark-bounded configuration, not a deployment approval. A real deployment requires corpus-validity checks, source display, monitored latency, privacy controls, abstention handling, and legal expert review.

8.5 Future Improvements

Priority work is (1) blinded legal-expert annotation for correctness, completeness, and claim-to-citation entailment; (2) seed-preserving graph and neighbor-budget sweeps; (3) BM25/tokenization and RRF-weight tuning under full-online latency measurement; (4) candidate-pool sweeps beyond C30; and (5) temporal-validity and external-reference evaluation. These studies should retain per-case candidates and provenance so failures can be audited.

8.6 Final Conclusion

The ablation evidence supports a practical conclusion: improve the ordering of available legal evidence with reranking before adding unvalidated retrieval complexity. Jina is the strongest observed reranker in the fixed-pool study, while the current hybrid, graph, and CoT variants do not improve their primary quality measures enough to offset their costs or risks. The result is an evidence-based engineering recommendation, bounded by the benchmark and not a claim of legal correctness.

Acknowledgments

Guidance: Acknowledge supervision, dataset preparation, software, and any external resources or funding.

References

Guidance: Add the cited scholarly, technical, dataset, and legal sources here after the substantive sections are completed.

A Dataset and JSONL Schemas

Guidance: Document the normalized document, provision, chunk, QA, evidence, and run-record schemas with representative JSONL fields.

B Example Documents, Provisions, Chunks, and Edges

Guidance: Provide compact, anonymized examples showing hierarchy, chunking, metadata, and graph-edge construction.

C Graph Relationship Vocabulary

Guidance: Define every relationship type, direction, source, confidence, and validity convention used by the knowledge graph.

D Retrieval and Reranker Configurations

Guidance: Record index settings, candidate budgets, fusion constants, reranker checkpoints, scoring conventions, and hardware-relevant settings.

E Prompt and Output Formats

Guidance: Include the base and reasoning prompts, expected structured outputs, citation format, refusal format, and parsing rules.

F Run Manifests and Additional Results

Guidance: Link or summarize run manifests, seeds, code snapshots, additional slices, latency summaries, and machine-readable result files.